

ACC Two-Channel Torque Assembly

Physical Conditions — Smoothstep Gate Inputs

Proximity $|\Delta x|$ · Quietness $\text{EMA}(|v|)$ · Stability θ ,
 θ · Contact Fz^L , Fz^R · Terrain h^L_{gnd} , h^R_{gnd}

Wheel Torque τ_w

τ_{balance} Sagittal Balance
 Core (pitch PD + vel. damp +
 pos-P + lateral + yaw)
 always active

+

$g_{\text{anchor}} \cdot \tau_{\text{anchor}}$
 Proximity-Gated Anchor
 (integral + damping boost)
 gated by g_{prox} , g_{env}

+

$g_{\text{flight}} \cdot \tau_{\text{flight}}$ Contact-
 Loss Recovery (reaction-
 wheel attitude PD) gated by
 g_{flight}

$\tau_w = \tau_{\text{bal}} + g_{\text{anc}} \cdot \tau_{\text{anc}} +$
 $g_{\text{flt}} \cdot \tau_{\text{flt}}$

Leg-Joint Torque τ_q

$\tau_{\text{posture}}(h_{\text{cmd}})$ Posture &
 Yaw Stability (Jacobian PD +
 hip-yaw divergence + homing)
 always active, height-
 scheduled

+

$g_{\text{terrain}} \cdot \Delta \tau_{\text{posture}}$ Per-
 Leg Ground Adaptation (split
 height commands + leveling
 integrator) gated by
 g_{terrain}



$\tau_q = \tau_{\text{post}} + g_{\text{terr}} \cdot \Delta \tau_{\text{post}}$

Smoothstep Gate Definitions

$g_{\text{prox}} = 1 - \text{ss}(|\Delta x|, 0.05, 0.15)$ $g_{\text{env}} = 1 - \text{ss}(\text{EMA}(|v|,$
 $0.18, 0.30))$

Asymmetric EMA: $\alpha_{\text{attack}} = 0.35$ ($\tau \approx 30\text{ms}$), $\alpha_{\text{release}} = 0.007$
 ($\tau \approx 1.5\text{s}$) $k_p = 50 - 15 \cdot (1 - g_{\text{env}} \cdot g_{\text{prox}})$

g_{flight} : $Fz < 0.5\text{mg}$ for ≥ 2 steps, release ≥ 5 steps +
 150ms ramp g_{terrain} : $|\Delta h| > 2\text{cm}$, unloaded freeze

Both channels assembled independently → final 10-DoF torque command

Wheel torque	Channel output	Gate definition
Leg-joint torque	Physical condition	Scheduled gain